CLASS 9 MATHEMATICS • CHAPTER 5

I'm Up and Down, and Round and Round

Exhaustive point-by-point presentation covering every concept, theorem, geometric visualization, and chord property directly from the textbook.

✦ Akash Srivastva (www.akashmaths.online)

Circles in Ancient Art & Nature's Patterns

Ancient Cave Art

Early humans were captivated by natural geometries:

- In ancient cave paintings, the Sun is universally depicted as a circle.
- **Gudahandi Cave Paintings (Odisha):** Feature numerous geometric patterns including triangles, squares, ovals, and circles inspired by nature.

Nature's Circular Ripple

Circles appear spontaneously in everyday natural phenomena:

- Expanding concentric circular waves when raindrops hit still water.
- Cross-section rings of a sliced tree trunk indicating annual growth rings.
- The geometric seed spiral inflorescence of a sunflower.

Celestial Circles

Skyward inspiration for ancient geometers:

- The full moon appears as a pristine glowing circular disc.
- During a total solar eclipse, the Moon aligns with the Sun, revealing the dazzling circular ring and glowing corona.

⚡ ⚡ QUICK BOX 1

CLASSROOM Q&A LIVE

Classroom Check: Nature & Circular Center

1 Which Indian historical cave site in Odisha is renowned for ancient geometric rock paintings of circles, triangles, and ovals?

Recall from textbook historical introduction.

2 Name three distinct physical occurrences in nature where circular shapes and patterns spontaneously appear.

Think about flora, weather, and astronomy.

3 **Jamuna's Puzzle:** If you are given an unlabelled circular paper cutout, what simple folding instruction can you give to pinpoint its exact center?

Hint: Think about lines of symmetry and creasing.

4 What fundamental geometric property is shared by every single circle, regardless of how small or large it is?

Focus on distance from a fixed point.

Mathematical Definition of a Circle & Locus

FORMAL DEFINITION & LOCUS

A **circle** is the set (or locus) of all points on a two-dimensional plane that are equidistant from a fixed given point on that plane.

Center & Radius

- **Center (A):** The fixed point from which all boundary points maintain equal distance.
- **Radius (r = AB):** The constant linear distance from the center to any point on the circle.

Chords & Diameters

- **Chord (BC):** A line segment joining any two distinct points lying on the circle.
- **Diameter:** A special chord that passes straight through the center. Its length equals 2r.

Subtended Central Angle

- **Angle at Center:** The angle formed when radii join the two endpoints of a chord to the center (e.g., $\angle BAC$).
- Every chord subtends a unique central angle.

⚡ ⚡ QUICK BOX 2

CLASSROOM Q&A LIVE

Classroom Check: Circle Anatomy & Locus

1. What is the mathematical definition of a "locus"? How does it apply to the creation of a circle?

   Express the condition in terms of a moving point.

2. Is every diameter a chord of the circle? Is every chord a diameter?

   Explain the distinction between the two terms.

3. If points B and C lie on a circle with center A, what type of triangle is $\triangle ABC$ always guaranteed to be?

   Consider the lengths of AB and AC.

4. Can a straight line intersect a circle at three distinct points on a flat plane? Why or why not?

   Think about collinearity and radii.

Infinite Rotational & Reflection Symmetries

Complete Rotational Symmetry

A circle possesses continuous, perfect rotational symmetry:

- A rolling vehicle wheel looks completely identical at every micro-angle of rotation.
- Unlike polygons that look identical only at discrete angles, a circle can be rotated about its center by **any angle θ** and remains unchanged.
- Order of rotational symmetry = **Infinite (Continuous)**.

Reflectional Symmetry Across Diameters

Folding a circular paper demonstrates its lines of symmetry:

- When folded so both semicircles overlap, the crease formed is always a **diameter** passing through the center.
- Every single diameter is an axis of reflection symmetry.
- Therefore, a circle possesses **infinitely many lines of symmetry**.

Comparison: Square (4 lines) • Regular Pentagon (5 lines) • Regular Hexagon (6 lines) • Circle (∞ lines)

⚡ ⚡ QUICK BOX 3

CLASSROOM Q&A LIVE

Classroom Check: Symmetries & Extreme Chords

1. In a circle with radius 5 units, what is the exact length of the longest possible chord?
   Think about the diameter.

2. Is there a "smallest" chord in a circle? What happens to chord length as endpoints approach each other?
   Consider the limit as endpoints get arbitrarily close.

3. What is the locus of all points on a plane that are equidistant from two fixed points A and B?
   Recall properties of perpendicular bisectors.

4. How many lines of reflection symmetry does a regular n-gon have compared to a circle?
   Contrast finite polygons with the infinite circular limit.

Circles Through Two Given Points

Locus of the Center

- If a circle passes through points A and B, its center O must satisfy: **OA = OB = radius**.
- The set of all points equidistant from A and B is the **perpendicular bisector** of segment AB.
- Therefore, the center of every circle passing through A and B lies somewhere on the perpendicular bisector of AB!

Count and Radius Variation

- **How many circles?** There are **infinitely many circles** passing through any two given points A and B.
- **Smallest Circle:** Has its center at the midpoint of AB; AB becomes its diameter. Radius = **½(AB)**.
- **Larger Circles:** As the center moves further up or down along the bisector, the radius increases indefinitely, and the circular arc through AB looks flatter and less curved!

Center Location: Any point P on perpendicular bisector of AB &implies; PA = PB (Valid Circle)

⚡ ⚡ QUICK BOX 4

CLASSROOM Q&A LIVE

Classroom Check: Circles Through Points

1 If the distance between points A and B is 10 cm, what is the radius of the smallest circle passing through both A and B?

Calculate ½ of segment AB.

2 Can you draw a circle of radius 4 cm that passes through two points 10 cm apart? Why or why not?

Compare radius with the minimum required radius.

3 As you choose circle centers further away along the perpendicular bisector, does the arc through A and B look more curved or less curved?

Observe curvature changes as radius grows.

4 How many squares can be drawn on a plane having two given points A and B as adjacent corners?

Consider orientations on either side of the segment.

Theorem 1: Uniqueness of Circumcircle

THEOREM 1

There is one and only one (unique) circle passing through three non-collinear points.

Why Is This True?

- Let the points be A, B, C (non-collinear).
- Center O must lie on perpendicular bisector of AB (OA = OB).
- Center O must also lie on perpendicular bisector of AC (OA = OC).
- Since A, B, C are non-collinear, these two lines intersect at a **unique single point O**.

Circumcenter Positions

- **Acute △:** Circumcenter O lies **inside** the triangle.
- **Right-angled △:** Circumcenter O lies exactly at the **midpoint of the hypotenuse**!
- **Obtuse △:** Circumcenter O lies **outside** the triangle.

Collinear Points Case

- If A, B, C are collinear (on one straight line), the perpendicular bisectors of AB and BC are **strictly parallel**.
- Parallel lines never meet on a plane.
- Hence, **no circle** can pass through three collinear points!

⚡ ⚡ QUICK BOX 5

CLASSROOM Q&A LIVE

Classroom Check: Circumcenter & Triangles

1 Where does the circumcenter of a right-angled triangle lie? What is the circumradius equal to?

Look at the hypotenuse relationship.

2 If a triangle has an obtuse angle of 110°, will its circumcenter fall inside or outside the triangle?

Recall Theorem 1 circumcenter variations.

3 Why is it impossible to construct a circle passing through three collinear points?

Refer to the behavior of perpendicular bisectors of parallel segments.

4 Can multiple congruent triangles share the exact same circumcircle?

Think about rotating or reflecting the triangle within the circle.

THEOREM 2 • EQUAL CHORDS

Theorem 2: Equal Chords & Central Angles

THEOREM 2

Equal chords of a circle subtend equal angles at the center of the circle.

Given & Objective

- **Given:** A circle with center C. Chords AB and DE are of equal length (AB = DE).
- **To Prove:** ∠ACB = ∠DCE.
- **Visual:** Imagine rotating a stretched thread of fixed length along the circumference of a wheel.

Proof via SSS Congruence

- In △CAB and △CDE:
- CA = CD = r (radii of the circle)
- CB = CE = r (radii of the circle)
- AB = DE (given: equal chord lengths)
- Therefore, **△CAB ≅ △CDE** by **SSS Congruence**!
- Hence, **∠ACB = ∠DCE** (CPCTC).

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⚡ ⚡ QUICK BOX 6

CLASSROOM Q&A LIVE

Classroom Check: Equal Chords & Radii Triangles

1 Show that the triangle formed by any chord and the center of the circle is always an isosceles triangle.

Which two sides are always guaranteed equal?

2 If chord AB = 8 cm subtends 64° at center C, what angle does another chord PQ = 8 cm subtend at center C?

Apply Theorem 2 directly.

3 Which triangle congruence criterion is utilized in the proof of Theorem 2?

Check the three pairs of equal corresponding sides.

4 If two chords have unequal lengths, will the longer chord subtend a larger or smaller angle at the center?

Think about opening your arms wider to grab a longer chord.

Theorem 3: Converse of Equal Central Angles

THEOREM 3

Chords of a circle that subtend equal angles at the center are equal in length.

Intuitive Physical Model

Imagine standing at center C with arms open at angle $a = 54°$:

- Your fingertips touch the circle at points A and B. The chord length is fixed by your arm span.
- Now rotate your body to a new position without altering the angle of your arms.
- Your fingertips now touch points E and D. The chord length ED must remain identical!

Proof via SAS Congruence

- **Given:** $\angle ACB = \angle DCE$.
- In $\triangle ACB$ and $\triangle DCE$:
- $CA = CD = r$ (radii)
- $\angle ACB = \angle DCE$ (given)
- $CB = CE = r$ (radii)
- Therefore, **$\triangle ACB \cong \triangle DCE$** by **SAS Congruence**!
- Hence, **$AB = ED$** (CPCTC).

⚡ ⚡ QUICK BOX 7

CLASSROOM Q&A LIVE

Classroom Check: Converse Reasoning

1. What is the key difference between the statements of Theorem 2 and Theorem 3?
Identify which is the condition and which is the conclusion.

2. Why does the proof of Theorem 3 use SAS congruence, while Theorem 2 uses SSS congruence?
Look at what information is initially provided in each.

3. If two central angles are 90° each in a circle of radius r, what is the length of each corresponding chord?
Use the Pythagorean theorem on the isosceles right triangle.

4. Two chords in congruent circles subtend equal angles at their respective centers. Are the chords equal?
Extend Theorem 3 across congruent circles.

Theorem 4: Center to Midpoint Line is Perpendicular

THEOREM 4

The line joining the center of a circle to the midpoint of a chord is perpendicular to the chord.

Geometric Setup

- Let C be the center of the circle.
- Let AB be a chord, and let M be its midpoint (AM = BM).
- Connect C to M, A, and B to construct △CMA and △CMB.
- **Objective:** Prove that ∠CMA = ∠CMB = 90°.

Step-by-Step Proof

- In △CAB, CA = CB = r &implies; △CAB is isosceles, so ∠A = ∠B.
- In △CMA and △CMB:
  - CA = CB (radii)
  - AM = BM (M is midpoint)
  - ∠A = ∠B (base angles of isosceles △)
- By **SAS (or SSS)** congruence: **△CMA ≅ △CMB**.
- &implies; ∠CMA = ∠CMB. Since they form a linear pair on AB:
- ∠CMA + ∠CMB = 180° &implies; **∠CMA = ∠CMB = 90° (CM ⊥ AB)**!

⚡ ⚡ QUICK BOX 8

CLASSROOM Q&A LIVE

Classroom Check: Midpoints & Altitudes

1 If △ABC is an isosceles triangle inscribed in a circle with AB = AC, show that the altitude from A to BC passes straight through the center.

Connect the altitude with the perpendicular bisector of BC.

2 Why do ∠CMA and ∠CMB form a linear pair in Theorem 4?

Because M lies directly on straight line segment AB.

3 Does the segment joining the center to any point on a chord make a 90° angle, or only when it joins the midpoint?

Consider what happens when the point moves away from M.

4 If chord AB has length 12 cm, what are the lengths of AM and MB when CM ⊥ AB?

Recall the definition of a midpoint.

Theorem 5: Perpendicular from Center Bisects Chord

THEOREM 5 (CONVERSE OF THEOREM 4)

The perpendicular dropped from the center of a circle to a chord of the circle bisects the chord.

Proof via RHS Congruence

- **Given:** $CM \perp AB$ (i.e., $\angle CMA = \angle CMB = 90°$).
- In right triangles $\triangle CMA$ and $\triangle CMB$:
  - Hypotenuse CA = Hypotenuse $CB = r$ (radii)
  - Common side $CM = CM$
  - $\angle CMA = \angle CMB = 90°$
- By **RHS Congruence**, **$\triangle CMA \cong \triangle CMB$**.
- Therefore, **$AM = BM$** (CPCTC). The chord is bisected!

Baudhāyana-Pythagoras Relation

Connecting center, midpoint, and endpoint creates a right triangle:

- Let radius = r, perpendicular distance = d, chord length = L.
- Half chord length: **$(L/2)^2 + d^2 = r^2$**

Chord Length $L = 2\sqrt{(r^2 - d^2)}$

⚡ ⚡ QUICK BOX 9

CLASSROOM Q&A LIVE

Classroom Check: Chord Bisector Applications

1 A chord of length 16 cm is at a perpendicular distance of 6 cm from the center. What is the radius of the circle?

Use $(16/2)^2 + 6^2 = r^2$.

2 Two parallel chords of lengths 6 cm and 8 cm are drawn on opposite sides of the center in a circle of radius 5 cm. What is the distance between them?

Find the two individual perpendicular distances d_1 *and* d_2.

3 If the two parallel chords from the previous question were on the SAME side of the center, what would the distance between them be?

Subtract the two perpendicular distances: $|d_1 - d_2|$.

4 How can you find the center of an arbitrary circle using only two non-parallel chords and a ruler/compass?

Draw their perpendicular bisectors and locate the intersection.

Theorem 6: Equal Chords Are Equidistant from Center

THEOREM 6

Chords of a circle having the same length are all at the same distance from the center of the circle.

Method 1: Congruent Triangle Altitudes

- Let chords AB = FG. Join center C to endpoints: △CAB and △CFG.
- CA = CF = r and CB = CG = r.
- Since AB = FG, **△CAB ≅ △CFG** by SSS congruence!
- Altitudes dropped to corresponding bases of congruent triangles must be equal.
- Hence, perpendicular distances **CE = CH**!

Method 2: Baudhāyana-Pythagoras

- E and H are midpoints &implies; AE = AB/2 and FH = FG/2.
- Since AB = FG, we have **AE = FH**.
- In right △CEA and △CHF:
  $CE^2 = CA^2 - AE^2 = r^2 - AE^2$
  $CH^2 = CF^2 - FH^2 = r^2 - FH^2$
- Since AE = FH, $CE^2 = CH^2$ &implies; **CE = CH**!

⚡ ⚡ QUICK BOX 10

CLASSROOM Q&A LIVE

Classroom Check: Equal Chords & Distances

1 If two chords in a circle each have length 10 cm, and one chord is 4 cm away from the center, what is the distance of the other chord from the center?

Apply Theorem 6 directly.

2 How does paper-folding with a tracing paper demonstrate that rotating equal chords preserves their distance from the center?

Describe the rotational symmetry of the circle.

3 If two equal chords intersect inside a circle, what geometric shape is traced out if you consider all chords of that identical length?

Hint: Midpoints stay at a fixed distance from the center!

4 In right $\triangle CEA$, if AE is 4 cm and radius CA is 5 cm, what is the perpendicular distance CE?

Recall the (3, 4, 5) Pythagorean triplet.

Theorem 7: Equidistant Chords Are Equal in Length

THEOREM 7 (CONVERSE OF THEOREM 6)

Chords of a circle that are equidistant from the center have equal length.

Algebraic Justification

- Let the perpendicular distances be CE and CH, with **CE = CH = d**.
- In right △CEA: $AE^2 = CA^2 - CE^2 = r^2 - d^2$.
- In right △CHF: $FH^2 = CF^2 - CH^2 = r^2 - d^2$.
- Since both equal $r^2 - d^2$, we have **AE = FH**.
- By Theorem 5, chords are bisected: AB = 2(AE) and FG = 2(FH).
- Therefore, **AB = FG**!

Geometric Insight: Concentric Locus

The collection of all chords of fixed length L in a circle of radius r:

- All their midpoints are at distance $d = \sqrt{r^2 - (L/2)^2}$ from the center.
- The locus of their midpoints forms a **concentric inner circle**!
- All these chords are tangent to this inner concentric circle.

⚡ ⚡ QUICK BOX 11

CLASSROOM Q&A LIVE

Classroom Check: Equidistant Chords & Midpoint Loci

1. Chords AB and CD are both 3 cm away from the center of a circle with radius 5 cm. What can you state about the lengths of AB and CD?

   Apply Theorem 7.

2. What is the geometric path (locus) traced by the midpoints of all chords of length 8 cm drawn in a circle of radius 5 cm?

   Identify the shape and center.

3. If the distance of a chord from the center is zero, what special chord is it, and what is its length?

   Consider a chord passing directly through the center.

4. If chords AB and AC are congruent, explain why the center of the circle must lie on the bisector of ∠BAC.

   Use congruent right triangles from the center to both chords.

Theorem 8: Longer Chords Are Closer to the Center

THEOREM 8

Between two unequal chords of a circle, the longer chord is closer to the center (has smaller perpendicular distance).

Proof via Pythagorean Invariant

- Let AB > DE. Drop perpendiculars $CF \perp AB$ and $CG \perp DE$.
- Radii are equal: $AC = CD = r$.
- By the Baudhāyana-Pythagoras theorem: $r^2 = CF^2 + AF^2 = CG^2 + GD^2$
- Since AB > DE, half lengths satisfy: **$AF > GD$** &implies; **$AF^2 > GD^2$**.
- Since the sum (r^2) is constant: **$CF^2 < CG^2$ &implies; $CF < CG$**!

Extremes & Boundaries

- **Closest Possible Chord:** The diameter passes right through the center ($d = 0$). It is the **longest chord** possible (length = $2r$).
- **Farthest Limit:** As a chord is pushed outward, its distance d approaches r, its length shrinks to 0, and it touches the circle as a single point (tangent).
- *Warning (Ex 5.5 Q3):* Distance and length are not linearly proportional! Doubling distance does NOT halve the chord length.

⚡ ⚡ QUICK BOX 12

CLASSROOM Q&A LIVE

Classroom Check: Unequal Chords & Distances

1 Between two chords of lengths 12 cm and 16 cm in the same circle, which chord lies closer to the center?

Apply Theorem 8.

2 If the distance of chord AB from the center is twice the distance of chord CD, does it mean CD = 2(AB)? Why or why not?

Check the square root formula $L = 2\sqrt{(r^2 - d^2)}$.

3 Why can no chord in a circle ever be strictly longer than its diameter?

Use triangle inequality on △AOB with radii OA and OB.

4 Through any given point A strictly inside a circle, what is the orientation of the shortest chord that passes through A?

Hint: It must be perpendicular to OA!

Arcs of a Circle & Angles Subtended at the Center

What Is an Arc?

- An **arc** is a connected portion of the circle between two boundary endpoints A and B.
- Two points A and B divide the circle into two arcs:
  - **Minor Arc (AXB):** The shorter path along the circumference.
  - **Major Arc (AYB):** The longer path along the circumference.

Central Angle Swept

- The angle subtended by an arc at center O is the measure of the angle swept as we rotate from OA to OB along the arc.
- **Minor arc subtends < 180°** at center O.
- **Major arc subtends > 180° (reflex angle)** at center O.
- A **semicircle** subtends exactly **180° (straight angle)**.

Inscribed Angle Invariance

- Angle subtended by arc AXB at a point C on the circle outside the arc means ∠ACB.
- Remarkable Discovery: The value of ∠ACB **does not change** regardless of which specific point C is picked on the alternate segment!

⚡ ⚡ QUICK BOX 13

CLASSROOM Q&A LIVE

Classroom Check: Arcs & Angles Swept

1 If an arc subtends a central angle of 130°, is it a minor arc, a semicircle, or a major arc?

Compare with 180°.

2 What is the central angle subtended by the major arc corresponding to a minor arc that subtends 70°?

Calculate 360° – 70°.

3 When do the minor arc and major arc have identical lengths? What is the central angle subtended in that case?

Think about dividing the circle into two halves.

4 If points P, Q, and R lie on the circle outside arc AB, what is the relationship between angles ∠APB, ∠AQB, and ∠ARB?

Recall the invariant inscribed angle property.

Theorem 9: Central Angle is Double the Inscribed Angle

THEOREM 9 (THE MASTER INSCRIBED ANGLE THEOREM)

The angle subtended by an arc at the center of the circle is double the angle subtended by the arc at any point on the circle outside the arc.

Proof Breakdown (Case 1: Center Inside)

- Join point D to center C and extend line DC to meet arc at E.
- $\triangle CDB$ is isosceles ($CB = CD = r$) &implies; $\angle CBD = \angle CDB$.
- Exterior angle of $\triangle CDB$: $\boldsymbol{\angle BCE = 2\angle BDC}$.
- Similarly in $\triangle CDA$ ($CA = CD = r$): $\boldsymbol{\angle ACE = 2\angle CDA}$.
- Adding both: $\angle BCA = \angle BCE + \angle ACE = 2(\angle BDC + \angle CDA) = \boldsymbol{2\angle BDA}$!

Universality (Case 2: Center Outside)

- When extension of DC falls outside the arc AFB:
- We subtract the angles instead of adding: $\angle ACE = 2\angle ADC$ and $\angle BCE = 2\angle BDC$
- $\angle ACB = \angle ACE - \angle BCE = 2(\angle ADC - \angle BDC) = \boldsymbol{2\angle ADB}$!
- Holds universally for minor arcs, semicircles, and major arcs (using reflex angle at center).

Central Angle $\angle ACB$ = 2 × Inscribed Angle $\angle ADB$

⚡ ⚡ QUICK BOX 14

CLASSROOM Q&A LIVE

Classroom Check: Theorem 9 Calculations

1 An arc subtends an angle of 70° at the center of a circle. What is the measure of the angle subtended by this arc at any point on the remaining circle?

Calculate 70° / 2.

2 If the angle subtended by an arc at a boundary point is 42°, what is the angle subtended by the same arc at the center?

Calculate 2 × 42°.

3 In the proof of Theorem 9, which foundational triangle theorem is used to relate interior angles to exterior angles?

Recall the Exterior Angle Theorem for triangles.

4 If central angle ∠AOB = 60° in a circle of radius 12 cm, what is the length of chord AB?

△AOB has OA = OB and vertex angle 60° &implies; equilateral △!

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Angles in Semicircles & Same Segment

Corollary 1: Angle in a Semicircle is 90°

Let AB be a diameter of the circle with center C:

- The diameter is a straight line, so it subtends a central angle of **∠ACB = 180°**.
- By Theorem 9, for any point D on the semicircle:
- **∠ADB = ½(∠ACB) = ½(180°) = 90°**!
- *"Angle subtended by a diameter at any point on the circle is always a right angle."* (Thales' Theorem)

Corollary 2: Angles in the Same Segment

Consider arc AB and points D, E, F in the alternate segment:

- Every point on the circle outside arc AB subtends half the constant central angle ∠ACB.
- Therefore: **∠ADB = ∠AEB = ∠AFB = ½(∠ACB)**.
- *"Angles subtended by the same arc in the same segment of a circle are equal to one another."*
- Points inside or outside the circle subtend strictly different angles!

⚡ ⚡ QUICK BOX 15

CLASSROOM Q&A LIVE

Classroom Check: Semicircles & Segments

1 If AB is a diameter and C is any point on the circle, what is the measure of ∠ACB?

Apply the semicircle corollary.

2 In Fig. 5.25, why are the angles ∠AIB and ∠AHB (points inside the circle) different from the angle ∠ADB on the circumference?

Connect to how angle changes when moving closer to the chord.

3 If points X and Y lie on the same side of chord AB on the circle, can ∠AXB ever be different from ∠AYB?

Refer to the constant angle in the same segment.

4 In a circle, a triangle is formed by the diameter and a point on the circumference. What kind of triangle is it always?

Think about its angle measures.

Theorem 10: Test for Four Concyclic Points

THEOREM 10

If a line segment AB subtends equal angles at two points C and D lying on the same side of AB, then the four points A, B, C, D lie on a circle (they are concyclic).

Setup & Proof by Contradiction

- **Given:** C and D are on the same side of AB with ∠ACB = ∠ADB.
- By Theorem 1, a unique circle passes through non-collinear points A, B, C.
- Suppose D does NOT lie on this circle. Then D must be either **outside** or **inside** the circle.

Elimination of Alternatives

- Let line AD intersect the circle at point E.
- Since C and E are on the circle in the same segment: **∠AEB = ∠ACB = ∠ADB**.
- If D is outside, ∠AEB is an exterior angle of △BED &implies; ∠AEB > ∠ADB (Contradiction!).
- If D is inside, ∠ADB is an exterior angle of △BED &implies; ∠ADB > ∠AEB (Contradiction!).
- Hence, D must lie precisely on the circle. **A, B, C, D are concyclic!**

⚡ ⚡ QUICK BOX 16

CLASSROOM Q&A LIVE

Classroom Check: Concyclicity Criteria

1 What is the special mathematical term used to describe points that all lie on the exact same circle?

Recall the definition from Section 5.8.

2 Line segment AB subtends 45° at point C and 45° at point D, both on the same side of AB. Do A, B, C, D lie on a circle?

Check Theorem 10 conditions.

3 Why does the assumption that D lies outside the circle lead to an impossible geometric contradiction?

Think about exterior angle vs interior opposite angle in $\triangle BED$.

4 If C and D were on OPPOSITE sides of AB with $\angle ACB = \angle ADB$, would A, B, C, D necessarily be concyclic?

Notice the requirement "lying on the same side of AB".

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Opposite Angles of a Cyclic Quadrilateral

THEOREMS 11 & 12 (DIRECT & CONVERSE)

A quadrilateral is cyclic if and only if the sum of either pair of its opposite angles is 180° (supplementary).

Theorem 11 Proof (Opposite Angles Sum = 180°)

- Let ABCD be a cyclic 4-gon with center O.
- Arc BCD subtends reflex angle BOD at center: **∠BAD = ½(reflex ∠BOD)**.
- Arc BAD subtends central angle BOD at center: **∠BCD = ½(∠BOD)**.
- Sum: ∠BAD + ∠BCD = ½(reflex ∠BOD + ∠BOD).
- Complete revolution = 360° &implies; **∠BAD + ∠BCD = ½(360°) = 180°**!

Theorem 12 (Converse Criterion)

- If in a quadrilateral ABCD, **∠A + ∠C = 180°**, then vertices A, B, C, D must lie on a circle.
- **Exterior Angle Property (Ex Q21):** In any cyclic quadrilateral, the exterior angle at any vertex equals the interior opposite angle! (e.g., ∠CDE = 180° – ∠ADC = ∠ABC).
- **Parallelograms:** The ONLY parallelogram that can be cyclic is a **rectangle**!

⚡ ⚡ QUICK BOX 17

CLASSROOM Q&A LIVE

Classroom Check: Cyclic Quadrilaterals

1 In cyclic quadrilateral ABCD, if ∠A = 75° and ∠B = 110°, what are the measures of ∠C and ∠D?

Subtract from 180°.

2 A quadrilateral has angles measuring 80°, 110°, 100°, and 70° in consecutive order. Can it be inscribed in a circle?

Check if opposite pairs add up to 180°: (80°+100°) and (110°+70°).

3 Why is a rectangle the only type of parallelogram that can ever be inscribed in a circle?

In a parallelogram opposite angles are equal, and in cyclic quad they add to 180°.

4 If a rhombus is inscribed in a circle, what special shape must that rhombus turn out to be?

A rhombus whose angles are 90° each.

Model Problem Solutions from Exercises

Problem 1: Chord Length (Ex Q1)

Given: Distance d = 5 cm, Radius r = 13 cm. Find chord length L.

- Drop perpendicular from center to bisect chord into two halves of length x.
- In right △: $x^2 + 5^2 = 13^2$
- $x^2 = 169 - 25 = 144$ &implies; x = 12 cm.
- Total Chord Length L = 2x = 2(12) = **24 cm**.

Problem 2: Arc Subtended (Ex Q2)

Given: Arc subtends 70° at the center. Find angle at circle boundary.

- Let central angle ∠AOB = 70°.
- Let P be any point on the alternate segment.
- By Theorem 9: ∠APB = ½(∠AOB).
- ∠APB = ½(70°) = **35°**.

Problem 3: Cyclic Quad (Ex Q8)

Given: Cyclic PQRS with ∠P = (2x + 10)° and ∠R = (3x – 20)°. Find x, ∠P, ∠R.

- Opposite angles are supplementary:
- (2x + 10) + (3x – 20) = 180°
- 5x – 10 = 180° &implies; 5x = 190° &implies; **x = 38°**.
- **∠P = 2(38) + 10 = 86°**
- **∠R = 3(38) – 20 = 94°** (86° + 94° = 180°).

Master Geometry Reference Checklist

● Centers, Chords & Symmetries

- ✅ **Locus Definition:** Set of all points equidistant from center.
- ✅ **Symmetry:** Infinite rotational and reflection symmetries across any diameter.
- ✅ **Circles Through Points:** ∞ through 2 points; exactly 1 unique circle through 3 non-collinear points (Circumcircle).
- ✅ **Theorems 2 & 3:** Equal chords ⇔ Equal angles at the center.
- ✅ **Theorems 4 & 5:** Center-to-midpoint line ⊥ chord ⇔ Perpendicular from center bisects chord.
- ✅ **Theorems 6 & 7:** Equal chords ⇔ Equidistant from center.

Arcs, Angles & Cyclic Quadrilaterals

- ✅ **Theorem 8:** Longer chord is closer to the center; Diameter is the longest chord.
- ✅ **Theorem 9:** Central angle is double the inscribed angle on remaining circumference.
- ✅ **Corollaries:** Angle in a semicircle = 90°; Angles in the same segment are equal.
- ✅ **Theorem 10 (Concyclicity):** Equal angles on same side of segment &implies; 4 points are concyclic.
- ✅ **Theorems 11 & 12:** Quadrilateral is cyclic ⇔ Opposite angles sum to 180°.
- ✅ **Rectangle Rule:** The only cyclic parallelogram is a rectangle.

END OF CHAPTER 5

THANK YOU

"The circle is the first, the simplest and the most perfect form — in it, the beginning and the end are one and the same."

✦ Akash Srivastva (www.akashmaths.online)